Evaluating a RAG-Based Chatbot for JRU CSE Program and Curriculum Inquiries

DecisionThree

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Abstract—This proposal outlines an exploratory evaluation of a retrieval-augmented generation (RAG) chatbot for graduating Grade 12 students considering José Rizal University's (JRU) College of Computer Studies and Engineering (CSE). The chatbot will answer inquiries about program focus, specializations, subjects, units, and documented curriculum requirements using the official College General Information, School Year 2025–2026, and relevant program profiles. The primary study will assess answer correctness and groundedness, defined as documentary support for factual claims. A supporting experiment will compare the chatbot with the same language model without retrieval using an identical, source-verified question set. Approximately 40 questions will cover fully answerable, partly answerable, and unsupported inquiries. Human reviewers will assess responses against reference answers and supporting passages. Local needs validation, source auditing, and a hardware pilot will precede evaluation. The project supports SDG 4 through access to educational information. No implementation results or performance improvements are claimed.

Keywords—retrieval-augmented generation, program inquiries, curriculum information, correctness, groundedness, SDG 4, JRU.

I. SELECTED UN SDG AND JUSTIFICATION

Selected SDG: SDG 4 - Quality Education. Target 4.3 concerns equal access to affordable and quality technical, vocational, and tertiary education [1]. Clear institutional information can help prospective students understand available educational options before enrollment.

Research on college-major information shows that an information intervention can influence students' choices [2]. This provides context for the project, but it does not establish a JRU-specific information-access problem. DecisionThree will conduct a short exploratory needs check before finalizing the inquiry categories.

The proposed chatbot will retrieve approved institutional passages and provide answers with source references. RAG combines retrieval with language generation [10]. The study will evaluate factual answer quality, rather than claim improvements in enrollment, affordability, or official SDG indicators. Demonstrating that students locate or understand information more easily would require a separate user evaluation.

II. NARROW PROBLEM STATEMENT AND CHATBOT FOCUS

Graduating Grade 12 students considering JRU CSE may have difficulty locating and understanding specific information about programs and curriculum requirements. Questions about program focus, required subjects, units, prerequisites, and internship requirements may require navigating lengthy institutional documents. The project will investigate whether a RAG-based inquiry chatbot can provide accurate answers supported by those documents.

Level	Group output
Broad domain	Access to tertiary-education information
SDG-related issue	Difficulty locating and understanding institutional program and curriculum information
Target users	Graduating Grade 12 students considering JRU CSE
Narrow problem	Obtaining specific, documented answers about CSE programs and curriculum requirements
Chatbot focus	Program and curriculum inquiries answered from approved JRU documents

Program coverage: The proposed scope includes Computer Engineering (BSCpE), Electronics Engineering (BSEcE), Information Technology (BSIT), the Business Analytics specialization (BSIT-BA), and Entertainment and Multimedia Computing (BSEMC), including Digital Animation Technology and Game Development [4]. Program and specialization labels will follow the approved source edition. BSEMC answers will distinguish its specializations whenever their requirements differ.

Included information: Program descriptions and specializations; curriculum subjects and units; documented course descriptions and prerequisites; and documented internship and project requirements. Coverage depends on available evidence. Missing prerequisites will not be interpreted as proof that a course has none, and a listed internship subject will not establish undocumented training hours.

Initial exclusions: Tuition quotations and payment concerns, application processing or status tracking, live class schedules, faculty availability, personalized program recommendations, and requirements for later curriculum editions without supporting documents. The chatbot will identify its document edition and avoid presenting historical information as a verified current requirement.

Example inquiries: "What undergraduate programs does JRU's CSE offer?"; "What is BS Information Technology about?"; "What subjects are listed in the first year of Computer Engineering?"; "How many units does this subject have?"; "What prerequisites are listed for this course?"; "Does Game Development include an internship?"; and "What does the Business Analytics specialization cover?" These are proposed questions, not evaluated outputs. Ambiguous inquiries will prompt clarification of the program, specialization, or course code.

III. RESEARCH QUESTIONS

A. Primary Research Question

How accurately and with what level of documentary support can a RAG-based chatbot answer graduating Grade 12 students' inquiries about JRU CSE programs and curriculum requirements?

Correctness concerns factual accuracy against verified reference answers. Groundedness means that the approved documents support the answer's factual claims. The primary question evaluates the proposed chatbot's ability to address the target users' information needs through documented answers.

B. Supporting Research Question

How do answer correctness and groundedness differ between the RAG-based chatbot and the same language model without retrieval augmentation?

The supporting question tests the contribution of retrieval while keeping the language model and test questions consistent. Improvement is an empirical possibility, rather than an assumed result. Comparison size will no longer serve as the experimental variable.

IV. INITIAL KNOWLEDGE-BASE DATA SOURCES

A. Primary Institutional Document

The principal proposed source is JRU's *College Division General Information, School Year 2025–2026* [3]. Its contents include CSE curricula and course descriptions. Only passages relevant to the defined inquiry scope will enter the knowledge base. Its school-year label will be recorded separately from any curriculum-version information. Public availability establishes access, but applicability to the intended student cohort still requires confirmation.

B. Relevant Program Profiles

The official CSE listing [4] and program profiles [5]–[9] will provide supplementary program context. The listed profiles date from April 2024 and will not automatically override the 2025–2026 document. The group will check dates, ownership, applicability, and any conflict before approving a source. Where discrepancies remain unresolved, the chatbot will identify the discrepancy or acknowledge that it cannot verify the requirement.

C. Source Audit and Retrieval Preparation

For each approved source, the group will log its title, owner, URL, edition, curriculum version where stated, applicable program or specialization, page numbers, access date, and any research-use conditions. Extracted tables will be checked against the original pages. No student records, grades, or personal academic information are needed.

The proposed extraction approach will preserve each course code, title, units, and prerequisite statement together where possible. Program and specialization labels will accompany every passage. Chunking will follow document sections and intact course records rather than arbitrary page cuts. Before testing, the group will choose and record the embedding model, index, retrieval method, chunk limits and overlap, number of retrieved passages, filtering policy, context budget, and citation format. These implementation settings remain provisional until the pilot and will then be frozen. Test questions and reference answers will stay outside the retrieval index.

V. INITIAL EVALUATION DATA PLAN

A. Needs Validation and Question Collection

A small, voluntary group of graduating Grade 12 students will complete an informal inquiry walkthrough using existing JRU materials. Staff or faculty may also identify frequently asked questions. This exploratory activity will check the local relevance of the proposed problem and inform question wording. The group will follow university procedures for participants who are minors and will avoid collecting names or grades. It will not present this activity as a representative survey.

B. Preliminary Question Set

The initial target is approximately 40 held-out questions. Thirty fully answerable questions will provisionally allocate five questions to each of six categories: offerings; program focus and specializations; curriculum subjects; course units; course descriptions and prerequisites; and internship and project requirements. Five partly answerable questions and five in-scope questions lacking documentary answers will test missing-information handling. Final allocation depends on source coverage. Out-of-scope requests will form a separate boundary check and will not replace unsupported in-scope questions.

Every question will have an answerability label, program or specialization, required reference facts, and exact supporting passages. A faculty member or program representative will review the reference key. The development and evaluation question sets will remain separate. The sample size reflects

manageable exploratory work, rather than a statistical power calculation.

C. Controlled Experimental Conditions

RAG condition: The selected language model will receive the question, common instructions, and passages retrieved from the approved corpus. It will provide page or source references for documentary claims and acknowledge unavailable evidence.

Without-retrieval condition: The same model will receive the same question and common instructions without supplied source passages, document access, or external tools. It may use its pretrained knowledge and acknowledge uncertainty. Common instructions will not require answering only from supplied documents, because this condition receives none. Source-citation requirements apply to the RAG condition.

The checkpoint, precision, mode, generation settings, answer-length limit, and question wording will remain fixed. Each inquiry will start a fresh session. The initial evaluation will produce one response per question per condition, giving about 80 responses. A seed will be fixed where supported. This exploratory design does not fully estimate variability across repeated generations. Any later repetitions will follow a declared protocol applied equally to both conditions.

D. Human Scoring and Analysis

Two group members will independently score both conditions' outputs using the faculty-reviewed key. Response order and condition identifiers will be masked where feasible, although visible citations may reveal the condition. Agreement will be calculated before the reviewers resolve disagreements. The group will retain the scoring decisions and evidence for reproducibility.

Primary metrics: Correctness will be the proportion of emitted, checkable factual claims that agree with the reference key, including the proper program, specialization, and course. Groundedness will be the proportion of substantive factual claims supported by approved documentary passages. Unsupported, contradicted, and supported claims will be labeled separately. Both conditions will be judged against the same documents. A baseline claim may be grounded even without a citation if the documents support it.

Supporting measures: Completeness will record the proportion of required reference facts covered. Citation support will check whether citations in the RAG response identify passages that support the associated claims. Reviewers will also record appropriate acknowledgment of missing information and unnecessary refusal of answerable questions. Responses with no factual claims will receive an undefined claim-based score, rather than automatic full credit; they will remain visible in refusal counts and completeness results. For wholly unsupported questions, completeness is not applicable.

Scores will be calculated per response and summarized by condition, inquiry category, and answerability. For matched questions with defined scores in both conditions, the group will report paired differences and their sample counts. It will also report all refusals and undefined scores so exclusions cannot conceal failures. Findings will remain exploratory. Error analysis will distinguish missing retrieved evidence, incorrect interpretation, program or version confusion, and output truncation. RAGAs and citation evaluation provide methodological context [11], [12]; manual review remains the primary assessment.

VI. POSSIBLE LANGUAGE MODEL AND JUSTIFICATION

Proposed model: Qwen3-8B in non-thinking mode remains the preliminary choice. Its official model card identifies an Apache 2.0 license and a configuration switch for non-thinking mode [13]. The group will pilot it on available university or group hardware before committing to deployment. A quantized checkpoint or a hosted deployment may be necessary, but both experimental conditions must use the same selected configuration. No fine-tuning is planned.

Expected role: The model will formulate answers from the inputs available in each condition. Retrieval, source labeling, and indexing will operate separately. The initial evaluation will use English inquiries. Filipino or mixed-language support would require an additional validated question set. The pilot will record memory use, response time, context limits, and any inference costs.

VII. GROUP REFLECTION ON MANAGEABILITY

DecisionThree consists of Luke Joaquin Bernardo, John Cyruz Patron, and Job Russel Enriquez. The project remains bounded to one university, one college, and documented program and curriculum inquiries. Responsibilities will cover source auditing and reference preparation, chatbot implementation and logging, and evaluation coordination. The group will agree on individual assignments, with two members independently scoring outputs and an external academic reviewer validating the answer key.

The planned sequence is needs validation and source auditing; a hardware and retrieval pilot; prototype development; reference-key review and configuration freeze; paired evaluation; and analysis and reporting. Milestone dates will follow the course schedule. Access to applicable documents, an available reviewer, and sufficient computing resources are the main dependencies. If a topic lacks verified evidence, its coverage will be narrowed and disclosed. This proposal describes future work. It does not claim completed implementation, measured improvement, or demonstrated benefits to students' decision-making.

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